

Exercise

Restaurant Automation System (RAS):

A restaurant owner wants to computerize his order processing, billing, and accounting activities. He also expects the computer to generate statistical report about sales of different items. A major goal of this computerization is to make supply ordering more accurate so that the problem of excess inventory is avoided as well as the problem of non-availability of ingredients required to satisfy orders for some popular items is minimized. The computer should maintain the prices of all the items and also support changing the prices by the manager. Whenever any item is sold, the sales clerk would enter the item code and the quantity sold. The computer should generate bills whenever food items are sold. Whenever ingredients are issued for preparation of food items, the data is to be entered into the computer. Purchase orders are generated on a daily basis, whenever the stock for any ingredient falls below a threshold value. The computer should calculate the threshold value for each item based on the average consumption of this ingredient for the past three days and assuming that a minimum of two days stock must be maintained for all ingredients. Whenever the ordered ingredients arrive, the invoice data regarding the quantity and price is entered. If sufficient cash balance is available, the computer should print cheques immediately against invoice. Monthly sales receipt and expenses data should be generated whenever the manager would request to see them. The computer should be able to print a menu card containing the menu items and their prices.

1. Prepare a requirements specification for the RAS system.
2. Perform SA/SD based-on your specification obtained in (1).

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1. Prepare a requirements specification for the RAS system.
[Req. Doc. As per the discussed template.](#)
2. Perform SA/SD based-on your specifaicon obtained in (1).
[DFD Level 0, 1, 2, Structure chart in sync with \(1\).](#)

Exercise

Supermarket automation software (SAS):

The manager of a supermarket wants us to develop an automation software. The supermarket stocks a set of items. Customers pick up their desired items from the different counters in required quantities. The customers present these items to the sales clerk. The sales clerk passes the items over a bar code reader and an automatic weighing scale and the data regarding the item type and the quantity get registered.

- SAS should at the end of a sales transaction print the bill containing the serial number of the sales transaction, the name of the item, code number, quantity, unit price, and item price. The bill should indicate the total amount payable.
 - SAS should maintain the inventory of the various items of the supermarket. The manager upon query should be able to see the inventory details. In order to support inventory management, the inventory of an item should be decreased whenever an item is sold. SAS should also support an option by which an employee can update the inventory when new supply arrives.
 - SAS should support printing the sales statistics for every item the supermarket deals with for any particular day or any particular period. The sales statistics should indicate the quantity of an item sold, the price realized, and the profit.
 - The manager of the supermarket should be able to change the the price at which an item is sold as the prices of the different items vary on a day-to-day basis.
1. Prepare a requirements specification for the SAS system.
 2. Perform SA/SD based-on your specifcaiton obtained in (1).